

PREPARING TO PERFORM



Brent Perkins proving he's worthy of *Proper Soul*, 5.14a, New River Gorge, WV.

📷 Dan Brayack

CHAPTER

12

"If there's only one thing I would like to say, this is: enjoy the process. Don't worry about the result. Climbing must be fun." — Marc LeMenestrel²

INTRODUCTION

Ten weeks in the gym. ARC sessions, countless hangboard workouts, a blur of campus rungs. Pints of sweat, perhaps a few ounces of blood, and sore and peeling fingers. The training has been difficult, even monotonous, but the results have been dramatic — at least on paper. Now it's time to test this newly improved machine on the real thing. Your body is finely tuned, and the only remaining variable is the effort applied to the next redpoint attempt. Or is it?

While the majority of this text — and the majority of your training time — has so far focused mostly on physical, technical, and mental preparation for difficult sport climbing, there are *many* other factors that contribute to (or inhibit) performance. Climbers tend to obsess over two primary components of performance, *ability* and *effort*, but it is essential to understand the other relevant factors as well, and learn to manipulate them in your favor. While attending to details such as weather conditions won't catapult a climber through the grades, ignoring them will certainly sabotage him, unnecessarily delaying progress and preventing him from attaining his potential. On meaningful, at-the-limit climbs, such details will be the difference between success and failure.

This chapter begins with a discussion of the *external* factors that affect an ascent, and how to influence them to facilitate success. These factors include environmental conditions and equipment. Next, route rehearsal strategies are discussed, including methods for maximizing physical and mental readiness prior to a performance. Finally, techniques are presented for effectively executing the performance process.

EXTERNAL FACTORS

External factors can have a tremendous impact on performance. Factors like weather may seem impossible to control, but with good planning, the climber can increase the odds of encountering favorable conditions. Other external factors are well within the climber's control, but are often ignored. The following techniques will stack the deck in your favor.

ENVIRONMENTAL CONDITIONS

Climbing performance is greatly affected by outdoor environmental conditions, including temperature, humidity, wind, and precipitation. The time of year and time of day are both important factors as they directly affect the temperature and humidity of the performance environment. Because of finger perspiration, friction between the rock and fingers is best when the rock is cold and the air is dry, but not so cold that it causes the fingers to go numb during the ascent. This ideal temperature varies considerably between climbers, so experiment with different conditions and note the temperature, humidity, and effect on performance. Many climbers will find that after a bone-chilling warm-up, the fingers can tolerate continued cold conditions without subsequent numbing. This is usually the ideal state for difficult redpoints. Plan the training season and any climbing trips to increase the chances of encountering good *sending temps* for the most important performance objectives.

*"Climbing is often best done in cold conditions, as cold as you can stand."
— Jerry Moffatt⁴*

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Certain regions — such as the American Southeast — suffer from extreme humidity most of the year, making the routes at these crags significantly harder during these periods, so plan trips to these areas for the fall and winter. Generally, across America, spring and fall are considered the ideal climbing seasons, but spring is much less stable than fall. Many of the Western Desert and Rocky Mountain crags experience high wind in the spring. This is beneficial in the late spring, keeping crags relatively cool, but a showstopper in early spring. Exposed routes like multipitch towers or protruding *arêtes* can become unclimbable in severe wind. Furthermore, crags that receive large amounts of winter rain or snowfall tend to seep in the spring. With a bit of research, it is usually possible to determine if a specific goal route is likely to be dry at a given time of year.

Temperature and humidity vary significantly within a single climbing day. Often crags are much colder in the morning, but more humid. Humidity is actually a function of temperature, so it will decrease throughout the day as temperature rises. A breeze can also reduce humidity.



Thoughtful planning will reduce the chances of visiting a crag when it's out of condition. © Mike Anderson

Sun exposure and shade are important considerations. Virtually all performance climbing should be done in the shade. Direct sunlight will significantly heat the rock, while also making it difficult to see certain features and chalk marks. When it is extremely cold it may be sensible to climb in direct sunlight; however, this radiant warming is unreliable because it is dependent on wind and cloud cover. Most routes receive shade at some point in the day. A key element of the planning process should be to ascertain the times of day when the goal route is shady, and thus when it should be rehearsed and climbed.

Note the changing parameters (temperature, humidity, wind, sun/shade) and identify the optimal times for performance attempts. Calculate backward from your target redpoint window to establish times to wake up, arrive at the crag, begin the warm-up, etc. For popular routes, budget extra time to wait in line.

EQUIPMENT

The tools and apparel used during a challenging ascent should be carefully considered. Shoes are the single most important piece of performance equipment. Despite what the ads say, no shoe is perfect for all terrain. Ideally, each climber will have a few different models of shoe that perform differently, and perhaps even a few different pairs of certain models that are sized differently or in differing states of decay. Extensively loved (aka “worn out”) shoes tend to be much more flexible and sensitive, allowing them to excel on low-angle slabs, or on steep terrain that requires toeing in and hooking. Vertical thin edging demands stiffness and precision, so new shoes with thick rubber and a tight fit are ideal.

Crack climbing presents its own list of shoe requirements, depending on the crack size and rock type. If the route has plentiful footholds, select a shoe to match them. Cracks with few footholds outside the crack will require foot jamming — use stiffer shoes for wider foot jams, and thin, sensitive shoes for thin cracks. Some climbers believe loose-fitting shoes minimize discomfort with extensive foot jamming. For difficult multipitch climbs, consider bringing two pairs of shoes, one that excels on face climbing and another that excels on cracks.

Regardless of shoe selection, each pair should be treated with care. Do not walk around in climbing shoes! Do not belay in climbing shoes! Clean the soles after each burn (simply wipe any debris off on a piece of cloth). In order to maintain fit and performance, avoid cramming shoes in the bottom of your pack. Do

Darkness at Noon?

Smith Rock is famous for its striking vertical walls and the brilliant edging routes that line them. These routes are notorious for their viciously sharp holds, so good redpoint conditions can make the difference between a successful ascent and a forced two-week layoff (to allow skin to heal). Many of the best climbs face southeast, so it's common to start late each day, in anticipation of ideal evening conditions. Patience is rewarded with tacky rock and fewer crowds, as the impetuous hang their heads late in the day, lamenting their throbbing fingertips.

This aspect of Smith climbing is woven into the fabric of Smith's history. The classic testpiece *Darkness at Noon* was so named because the route gets shade precisely at noon. Climbers flock to this spot at noon each day, waiting as long as possible for the rock to cool off, but not so long that someone sneaks in first, not unlike the boarding line for a Southwest Airlines flight. Of course, you will need to adjust your alarm clock for the summer months; during daylight saving time, it's *Darkness at One*.



not leave shoes in the sun or near other heat sources. This goes for between burns as well. On especially cold days it may be prudent to zip shoes inside your down jacket to keep them warm, but most of the time shoes perform better cold, so find a shady alcove to store shoes between burns.

Extensive walking and high heat cause feet to swell, so avoid walking around too much on climbing days that require the use of tight performance shoes. If climbing in the sun in August, consider up-sizing to a worn-out pair of shoes. Generally, it's a good idea to bring at least two pairs to the crag in case something happens to the primary pair or they're just not working well for whatever reason. A pair of older or needing-to-be-broken-in shoes can be used for warming up to spare the wear and tear on the primary pair. Warm-up shoes should still fit and perform well, as discussed in **Chapter 3: Skill Development**.

The climber's harness should be well-fitting and built for the task at hand. A wall harness features many bells and whistles that add weight but aren't needed for a simple redpoint attempt. Use a harness with sufficient padding to provide comfort throughout a thorough working burn. Remove any extraneous objects like belay devices, locking carabiners, or belay-certification cards. Carry a small toothbrush at all times to clean holds and remove tick marks. A small

roll of tape should be carried on reconnaissance burns for preventative taping.

The climber's rope is a matter of preference, but the bottom line is to select a cord that inspires confidence, allowing the climber to commit 100 percent to difficult or runout moves. There are numerous lightweight varieties on the market, but these are more prone to slipping through some belay devices. Ensure that the rope is compatible with the belay device, and that the belayer can provide a safe catch. Thicker (and so, heavier) ropes are generally more abrasion resistant. Keep the rope clean to prevent contaminants from greasing up the hands.

Quickdraws can also affect performance. Use draws that are easy to clip, such as a wire-gate or bent-gate carabiner on the rope end of the nylon sling, secured to the nylon with rubber or tape so it cannot accidentally rotate into an unclippable position. Try different quickdraw configurations to determine which you prefer, then practice with those draws. Many climbers take clipping for granted, but it is not trivial. If clipping is a weakness, practice clipping while vegging out in front of the flat screen: Hold the bolt-end biner of your preferred style of quickdraw at arm's length, and then clip a length of rope into the rope-end biner 100 times with your other hand. Now rotate the quickdraw so the clipping gate horizontally faces the

opposite direction. Clip 100 times. Now switch arms and repeat. Move the quickdraw around in space to increase the difficulty of this drill. Usually only a few repetitions will cure even the most flustered clippers.

Considering the amount of money spent on clothing, it's amazing how little emphasis is placed on its performance qualities. Frankly, the rock jocks of the 1980s had it right: The ideal outfit is stretchy Lycra on the bottom and a loose tank top up above. Unfortunately, if men were to dress this way now, nobody would ever take their picture or kiss them (women, on the other hand, will be kissed more than they want if they dress this way), so a compromise is essential. Wear loose-fitting pants, preferably with stretchable fabrics, that shroud your knobby knees and tan lines yet do not inhibit flexibility. Leave keys, wallets, and smartphones in the backpack; they won't be needed at the fourth bolt. Some form of belt is usually helpful, but keep weight and bulk in mind and go with an integrated design or a simple piece of cord.

Upper-body clothing choices are purely personal. A shirt of some type is usually a good idea because it protects skin from rope burns and other abrasions, while providing a place to wipe sweat off the brow. Avoid tight-fitting shirts that can restrict blood flow or even pinch nerves in extreme cases. Bulky jackets or sweatshirts are almost always a bad idea. In extreme cold, consider wearing two or three T-shirts, or a vest. Most climbers will warm right up soon after they start climbing, but if it's really cold it may help to jog uphill for a few minutes just before tying in and then ditching any unneeded outerwear.

A hat or beanie should always be on hand. When in doubt, start up the climb with a beanie; it can be easily discarded once warmed up. Bracelets, necklaces, rings, etc., present a significant safety hazard so should never be worn while climbing.

ROUTE REHEARSAL

Route rehearsal is a significant part of the redpointing process that comes into play on every route that is not climbed successfully on the first attempt. The process of identifying the most effective sequence, internalizing this information, and preparing the mind to execute it under pressure is critical to realizing your climbing dreams. Mastering these skills takes practice and experience, but the following strategies will help jump-start the learning process.

"To climb hard routes, it's better to try every conceivable option...It'll save you time in the long run."

— Ethan Pringle⁵

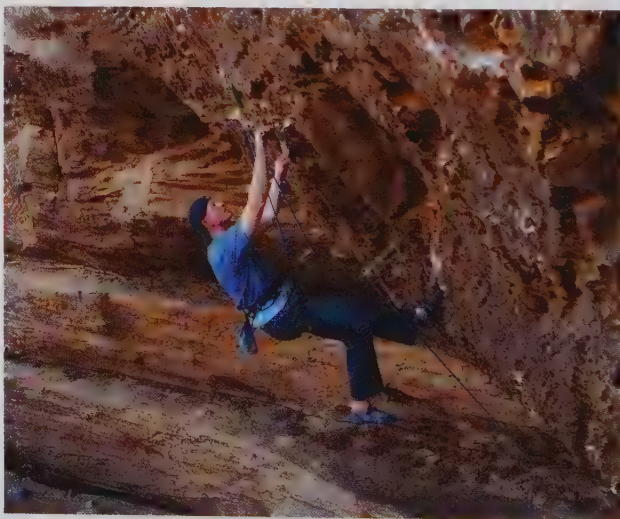
DISCOVERING AND REFINING BETA

Determining the optimal sequence of moves on a difficult route is not trivial and often involves a bit of luck. However, there are tactics that can be applied to virtually any climb to expedite this process. These methods will help identify an effective sequence in short order, facilitating a quick ascent and saving time and energy for other objectives.

POSITIONING FOR OPTIMAL LEARNING

The first tactic is to utilize the modern conveniences of sport climbing that our Lycra-clad forefathers fought and dieted for. Work out sequences while hanging from the rope, and ensure that the rope is in the optimal position for working those moves. For slabby to vertical routes, that might mean a top rope. On steeper terrain, the rope should usually be clipped to the next bolt above the climber (though on terrain steeper than 45 degrees, the next bolt at or below the climber may be better). If necessary, use a quickdraw or carabiner to clip (aka *tram*) your belay loop into the belayer's side of the rope to stay close to the rock. As progress is made up the route, continually reposition the rope clip-in point to enable optimal learning.

A stick clip is an essential tool for rope management during the beta-discovery process. The stick should be used not only to keep the climber safe, but also to aid in positioning the rope. If unable to climb



On steeper terrain, clip a quickdraw from your belay loop to the belayer's end of the rope to stay within reach of the rock. 📷 Mike Anderson

onsight from bolt-to-bolt on the first go, clip in directly to the nearest bolt and use the stick clip to advance the lead rope to the next higher bolt. This is especially helpful for exploring dynamic or blind sequences, which can cause unnecessary skin wear. Instead, by advancing the rope above the crux section, each hold can be closely inspected to find the best grip positions and aim points, or to decide if tape or other preventive measures are required to avoid a skin injury.



Like a samurai with his katana sword, Rain Murphy uses his stickclip to slice up his project. Dan Brayack

and each limb. This method can be used to brainstorm movement options, and will preserve the skin and muscles for the most promising possibilities.

ASSESSING THE OPTIONS

Actually unlocking the ideal sequence can be difficult, and the solution will vary greatly between routes. Think of every route as a series of boulder problems, and tackle each problem individually before considering the route as a whole. Use these steps to solve each individual problem:

- Identify all the potential holds and consider the useful attributes of each hold individually. Once the inventory is complete, consider how lesser holds might be used together (for example, two sidepulls in opposition).
- When examining potential handholds, consider every possible way the hold might be gripped. A hold may be used multiple times — perhaps as a gaston, and then later as a sidepull or undercling. Closely inspect potential footholds at eye level, identifying positive surfaces that can be hooked or used to pull the lower body into the rock.

THE BELAYER COACH

An engaged belayer can be extremely helpful, acting as a surrogate coach. In addition to providing moral support, a soft catch, and encouragement, an attentive belayer will note body positions during the execution of a difficult move and can help remind the climber of the successful sequence. Sometimes a third-person perspective can help identify potential options that the climber did not consider. The belayer should pay particular attention to the feet, as the climber rarely forgets which handholds were used. The climber and belayer should keep a running dialogue during this process to keep both parties on the same page and to help reinforce memorization of the correct sequence. An engaged belayer will learn from this sequence problem-solving as well (even if she has no immediate intention of climbing the route), improving her ability to read sequences and decipher cruxes. Keep this in mind when you are belaying.

In addition to utilizing the belayer's wisdom and perspective, it can help for the climber to imagine himself completing the move from a third-person perspective. The climber should attempt each move in his mind's eye before attempting it on the rock, including visualizing the movement of the hips, shoulders,



Clues in the rock: a thumb print to the right of a hold indicates a left handhold and vice versa. Black marks or tick marks with no other chalk indicate popular footholds. Mike Anderson

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- Assess the distances between the largest holds, and consider the pros and cons of making larger moves between the best holds versus smaller moves between lesser intermediate holds (or a combination of the two). Ace French climber Tony Lamiche advises to never underestimate the value of a big hold — even if it seems to be facing the “wrong” direction.
- Leftover chalk and rubber marks can provide clues to the most popular sequence (thumbprints below holds can hint at whether to use the right or left hand on a given hold), but don’t assume previous climbers discovered the most efficient sequence. If it’s possible to determine the correct hand for a given finishing hold, it’s often easier to work backwards from that point. Often the key to a particularly puzzling sequence is a “shuffle” move where both hands use a single hold (though not at the same time, as when matching hands).
- Lynn Hill suggests that footholds exist wherever they are needed — simply determine the correct location for your foot, and then place it there. The rubber will do the rest.

Consider every possible option for completing a difficult move, even if it’s already been climbed successfully. Often the first successful sequence is *not* the most efficient. Subtle changes in foot position or hip movement can make significant differences, so leave no stone unturned. If crux sections were worked when relatively fresh, return to them once you are fatigued to ensure your chosen sequence is executable when you are pumped or powered-down. Time spent dialing the moves is almost *never* wasted time. An extra 10 minutes of poking around at the crux can save days of unneeded redpoint attempts.

REST

Rest stances should be identified and evaluated. Any rest that the climber can sustain below the maximum steady state can be used indefinitely, as the climber should not incur more fatigue by remaining there (identifying this point comes from practice during base-fitness training). A poor rest may require substantial concentration to maximize weight on the feet while minimizing forearm grip effort. Some rests, such as kneebars or reachy stems, place physical demands on other parts of the body, and may become counterproductive after several minutes. Practice each rest during working burns — many climbers make the mistake of resting on the rope too much during working

burns, missing the opportunity to rehearse rest stances. Next, develop a plan for which rests will be used and for how long. It is wise to remain at very good rests for one to two minutes *after* you have completely relaxed, de-pumped, and your breathing and heart rate have settled down.

CLIPPING

Once the general climbing sequence is solved, consider each clip. Many will be trivial, but specific clips before, during, or after crux sections can be decisive. Position each quickdraw so that the rope-end carabiner is positioned for easiest clipping, keeping in mind also that your biner should face *away* from your direction of travel, especially on traverses. Identify the most efficient body position for clipping, which may be above or below the existing bolt placement. Extended draws can be used to allow clipping from a good stance below a crux. Sometimes a particular clip can be the hardest move on the route, and such clips are often skipped (in fact, certain mid-crux “dogging bolts” were placed with the intention that they would be skipped during redpoint attempts). Other times a clip is skipped just to save energy on a pumpy route. Whatever the rea-



Extend the length of fixed quickdraws so they can be clipped from the most efficient position, or to minimize rope drag.  Mike Anderson

son, keep your belayer in the loop when considering skipping clips. Discuss the strategy for difficult clips and the pros and cons of skipping clips to ensure that your belayer can provide a safe catch in the event of a fall. Consider taking a few shorter practice falls to evaluate the consequence of a fall before committing to a longer runout.

LOWERING

Once at the top of the route, examine it closely while lowering slowly. Note any areas that are still unclear, and try to recall the sequence for each section of the route (going from bolt to bolt or rest stance to rest stance). Pause while lowering if necessary. Once on the ground, discuss the beta and the quality of your attempt with your belayer.

LINKING THE ROUTE

Once the individual boulder problems are known, it's time to put them together. Often the easiest way to get from point A to point B makes it significantly more difficult to get from point B to point C. Reconcile these

conflicts by identifying the most efficient sequence between obvious rest points or clipping stances. As soon as you begin to link individual sections, identify any moves or sequences that require further beta refinement. Moves that require extensive grunting, inhibit breathing, or leave one of the four limbs unused are ripe for further analysis. Once the route can be climbed with only three or four hangs, consider the proper pacing for the route. This varies widely depending on the climber and the characteristics of the climb (including length, angle, relative move difficulty, and availability of rests, among other factors). Also, this pacing will change for different segments. Generally, it's wise to climb as quickly as possible without fumbling a sequencing or becoming frantic. Some pure power-endurance sequences are best climbed as an all-out sprint, while technical endurance sequences may best be climbed slowly, with precision, taking lots of rest between moves. Finally, remember to place extra emphasis on dialing the sequences at the end of the route, or other places where a building pump or anxiety will hinder movement execution.

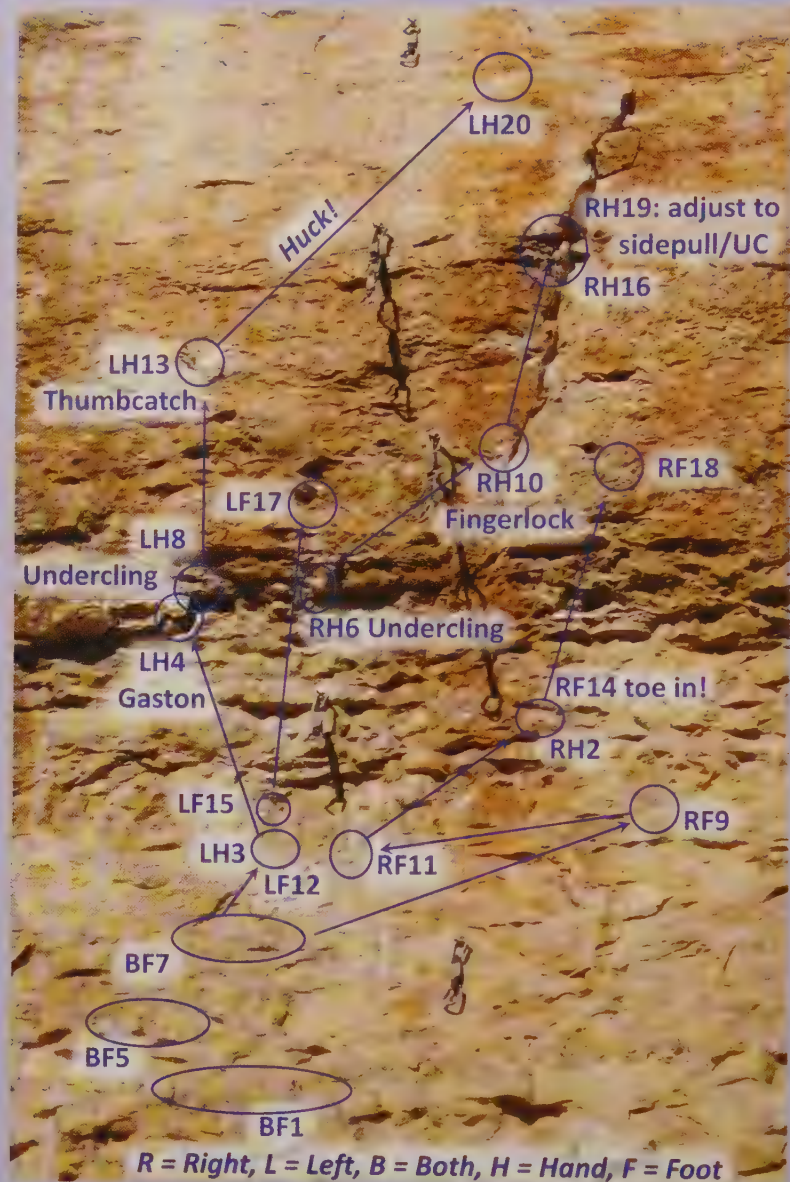


The Beta

There are many ways to document route beta, including videotaping the sequence to review later — a blow-by-blow record of the route's moves that inspired the coining of the term "beta," for the now-defunct Betamax videotape format. Modern point-and-shoot cameras and phones, coupled with a cheap tripod, make it easy for anyone to film a climbing sequence. The goal is not Oscar-worthy cinematography, but rather a broad, stable shot that captures the sequences in a consistent manner so that one day's sequence can be easily compared to the next.

In addition to video, it's a good idea to create a beta sheet or beta map. These tools can be reviewed at the crag before an ascent, or even carried on the route during a red-point burn (only review them after a fall!). Beta maps provide a visual representation of the route, and can be hand-drawn, or based on a photo like the one shown. Photo-based maps often capture the spacing and orientation of holds more accurately than hand-drawn maps.

In my experience, text-based beta sheets are the most effective, because they force the climber to think critically about every single move, and insert key reminders about effort, pacing, hip position, and other abstract or subtle factors that affect the ascent. A detailed beta sheet will describe how to grip each hold, when to clip, how to generate momentum, and when to relax. Below is an example of a detailed description of the opening moves of a different route (than the one pictured above). Note the details on hand position and pacing:



"Begin at Horizontal Break (HB), RF up on HB, RH to undercling/pinch diagonal plate along crack, cross LH over and MRP crimp diagonal edge just above slopey pod. RH go out right, avoiding rope, to crimp. Match LF to spike on HB, flag RF, LH to vertical slot, w/ pinkie in constriction. RF to highest part of diagonal ramp, then bump RH to Sidepull undercling (furthest right part, more of a sidepull than undercling). LF to triangular pod, then dyno/karate chop LH to next pod, walk LF, then RF, then LF to good chip out left, slap RH up to bad slopey sidepull, RF toe on spike at L edge of crimp, then pop RH up to gaston, stand a little & LH to mono. Don't rest, just keep trucking..."

*RH = right hand, LF = left foot, MRP = middle, ring, pinkie fingers, etc

— Mark Anderson

REMEMBERING THE BETA

The perfect sequence is useless if it's forgotten. Ascents of limit routes require exhaustive memorization of the entire sequence. This includes not only which holds to use, but exactly how to use them, how to move the hips and shoulders, and the timing of dynamic lunges. There are many ways to help this memorization process.

Key locations or hard-to-see target holds can be highlighted with subtle chalk marks, but be judicious, and thoroughly brush them away after each session. Alternatively, get in the habit of noting subtle features in the rock itself that can be used as reminders of key holds. It's important to note the location of hard-to-see holds, but also to note exactly how to position the hands and feet on every hold. Sometimes success hinges on securing a crux hold exactly right, so note any peculiarities in the feel of the hold's surface for future reference — for example, that minute spike that sticks up between the middle and ring finger-pads when a certain edge is held correctly.

As soon as possible after the ascent, write down the beta. Consider creating a visually descriptive *beta map*, with a rough sketch of the features of the route, indicating the correct sequence for moving between holds. These maps are popular, but often result in confusing spaghetti lines that are hard to follow. Beta maps are suitable for visual learners, while verbal learners may prefer move-by-move text descriptions which can be recalled as an internal dialogue. Text descriptions also allow for unlimited detail and capture subtleties like hip shifts or the perfect hand placement on a key hold. If possible, consider shooting simple video of the climbing for later review — it certainly doesn't hurt to use all three tools.

The beta map, beta sheet, and/or video should be reviewed regularly (at least three or four times per day) between climbing days. It's not uncommon for climbers to conjure up potential alternative sequences during the review process. Jot these down for future evaluation, but stick with the tested beta for the redpoint go, and then try the new option if you fall, once hanging from the rope.

"I remember all the moves of every problem I've ever done. This is what keeps me up at night."

— Dave Graham²



Some holds require precise placement of each individual finger. Include this information, and other subtle beta, when documenting your sequence.

VISUALIZATION

Visualization is a technique athletes have used for decades to improve confidence and motivation. In the case of choreographed performances, like a gymnastics routine or rehearsed redpoint attempt, visualization is also a powerful method of memorizing the sequence of moves. It provides opportunities for the mind to rehearse the moves when the body is physically separated from the project, or too worn out for additional hands-on repetitions.

With visualization, the climber imagines successfully climbing her project in her "mind's eye." This process can help eliminate doubts in the climber's mind, opening it to the possibility of success. Confidence is essential to any successful athletic performance, but it's even more important in redpoint climbing, because it helps maintain motivation over the course of an extended project. It's difficult to commit complete-

Discovering and Refining Beta: Key Takeaways

- Use a top rope or stick clip to position the rope for optimal learning
- Keep chalk marks to a minimum
- Get your belayer involved
- Consider all available holds
 - Consider holds in combination
 - Get the most out of the best holds
 - Footholds are everywhere
- Note clipping positions
- Review the beta while lowering and between burns
- Reconsider moves that require extensive grunting, inhibit breathing, or leave one of the four limbs unused
- Document the beta

ly to a challenging goal when the result is constantly in doubt. Imagining what success will ultimately “look” and “feel” like provides a bridge for the mind to transition from a state of skepticism to one of sincere belief. This realization that the goal (which often seems inconceivable at the outset) is truly within grasp can fuel the final surge of effort necessary to succeed.

With practice, visualization can be performed anywhere, but first-time practitioners should set aside a short block of time free from noise and distractions. Sitting or lying down in a darkened room is a great place to start. With eyes closed, imagine the goal route. Some find it easier to imagine a *third-person perspective*, envisioning their performance as though they were a member of the audience rather than the leading character. Ultimately, strive to imagine events as they will eventually be observed in real life: from a *first-person perspective*. While visualizing, breathe deeply, as if on the climb. Establish the setting, including the crag scene, the weather, and the audience. Imagine stepping through the typical preparations: warming up, miming the moves below the route, tying in and booting up. Imagine a safe, secure belay from a trusted partner.

Imagine starting up the route, and then climbing each individual move. Envision every single hand and foot movement, as well as key hip or shoulder movements. Initially the hand moves are easiest to remember, but on more challenging projects internalizing the foot beta will eventually be essential. Whenever possible, mime each move as you progress up the route in your mind. Imagine all the subtle intricacies of the route—particular ways of grasping uneven handholds, hip shifts, variations in pace, clips, rest, and relaxation points. Imagine battling through a difficult crux, including shouts or grunts if these will likely be part of the final performance. Imagine a building pump, but fighting that extra bit harder to reach the next rest. Finally, imagine clipping the rope into the anchor after the send, and the euphoric feelings that accompany success.

A common mistake is to imagine “floating” up the route with ease. Occasionally, this does indeed happen, but it is rare. More often, successful performances are extremely difficult, requiring tremendous effort. Expect to struggle, but imagine trying very hard and overcoming these difficulties. Expect to have moments of doubt, but imagine reassuring yourself with positive self-talk. Embracing these challenges and preparing for them, rather than ignoring them, will provide comfort at the moment of truth. You knew it

would be hard, so prepare to try hard.

Once the method is learned, visualization should be performed regularly throughout a project campaign. Take advantage of peaceful moments (such as when riding in a car or on a walk) to visualize. When time is limited, imagine the performance in sections. If possible, sit below the route while visualizing, to help you recollect any forgotten sequences. Video of the route can also be used to stimulate the visualization process, but the third-person perspective is not ideal. Use video to help refine the beta early in the campaign, but also take time for old-fashioned imagination.

CONFIDENCE

The expectations that a climber harbors about her potential ability to climb a given route contribute significantly to determining if she will. While this concept of the *self-fulfilling prophecy* is a cliché in many sports, in climbing it is very real. The climber’s attitude directly affects her physical execution in several critical ways, including breathing, physical relaxation, grip control, and movement analysis and execution. With the right



A broad base of climbing experience will help breed the confidence needed to succeed on your limit routes. Megan Howk boosting her pyramid on *Bare Metal Teen*, 5.12a, Red River Gorge, KY. © Dan Brayack

attitude, the climber's mind becomes a tremendous asset, able to perform the very demanding task of properly controlling these critical functions. With the wrong attitude, the mind will restrain an otherwise capable body. It becomes a burden that must be hauled up the route, tugging on the climber with every move.

A confident climber can relax and climb efficiently because she expects to overcome the challenges that arise, rather than bracing for retreat. Confidence that her feet won't slip allows her to relax her grip; confidence that the next hold will be adequate allows more-flowing movement; confidence that she will recover allows her to commit decisively to a crux; and so on. Achieving this mindset is fundamental to onsighting, so it is discussed further in **Chapter 13: Redpoint and Onsight Climbing**. However, confidence also has significant and distinct implications for redpoint climbing. During the initial route-rehearsal process, the climber must believe the send is possible eventually; otherwise her doubts will prevent maximum effort on the hardest moves. The mind will magnify these doubts to convince the climber that the moves are "impossible" and that such a lofty goal is a waste of time.

The confidence needed to combat the mind's meddling must spring from actual ability and achievements. Therefore, confidence must be cultivated and nurtured over a career. A *route pyramid* (described in **Chapter 2: Goal Setting and Planning**) is the perfect tool for building the broad base of experience needed to instill confidence for future projects. It helps identify chinks in the climber's armor, focusing her training effort to solidify her skills. Thus, by building diverse and thorough pyramids, the climber will have indisputable evidence that she is capable of achieving the next goal. It can be helpful to review these pyramids occasionally to reinforce this belief, especially at the outset of a new project or during moments of doubt. If she has done the work needed to complete the pyramid, and the pyramid leads logically to the goal route, then the climber has no reason to fear the route or doubt her ability to succeed. Instead, she should believe that she will find a solution and succeed on the route eventually — the proof lies in the stone foundation that is her pyramid.

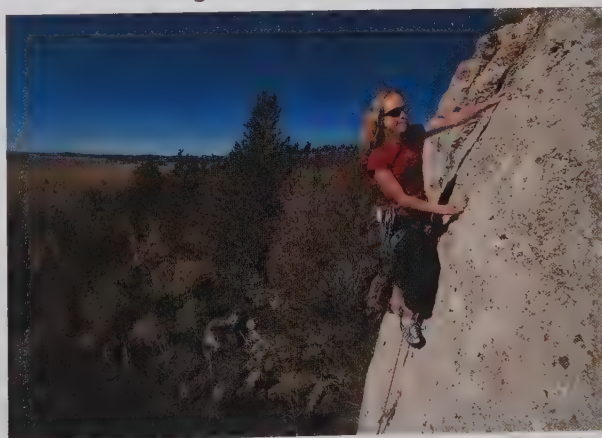
THE PERFORMANCE PROCESS

There are several proven methods for preparing the body and mind to perform at their best when it counts: on the most cherished goal routes. In climbing, each route is unique, and so is each performance. However, the same basic process can be used to prepare for any climbing performance. This process can and should be tailored to each climber and the demands of each route, but the fundamental components remain universal. Once a suitable routine is established, repeating it will provide comfort on high-anxiety performance days.

WARMING UP

A thorough warm-up is critical to any performance — it can literally make or break a climbing day. The best warm-up for a given climbing day can vary dramatically depending on the performance objective, but there are some common characteristics. An effective warm-up will increase circulation throughout the body, clear out any residual waste products in the muscles, raise the heart rate and breathing, and activate the neurologic processes that control motor movement.

Warm-up activities should begin relatively easily, and then gradually increase in difficulty until a moderate pump is achieved. Err on the side of selecting warm-up climbs that are too easy in order to avoid the dreaded flash pump. The warm-up climbs should depend on the intended activity for the day, the climber's skill, and her familiarity with the warm-ups, so grades are not a good indication of suitability. Duration is a better indicator. Thirty to 45 minutes of on-route climbing time should be set-aside for warm-



A complete warm-up is mandatory. Kate Anderson preparing to crush at Wild Iris, WY.  Mike Anderson

ing-up, which can include as many as four warm-up burns for skilled climbers at a familiar crag. Don't rush through the warm-ups either, especially if climbing familiar routes; climb at a pace similar to the goal route. Try to select routes that are sustained, have comfortable holds, and are enjoyable to climb, so that they elicit a positive mood. Most popular crags have well-traveled warm-ups that fit the bill — ask locals for guidance and arrive early to get in the queue. If the best available warm-up is too easy, skip a few holds or linger on marginal holds to achieve the desired intensity.

Include some range-of-motion activities and active stretching while on route. Warm-up climbs should be done on lead to warm-up the “lead head” as well. Breathe steadily and intently throughout the warm-up to habituate the best behavior for later performances. The nervous system requires a thorough warm-up as well, in order to optimize balance, motor control, and muscular recruitment. Disproportionally powerful efforts require a more intense warm-up to get all of the muscles firing effectively. Once the warm-up is complete, allow ample time to recover before moving on to the day's objective (30 to 60 minutes of rest is common). On occasion, it may be necessary to re-warm-up at times throughout the day if too much time passes between burns. One moderate pitch should suffice.

“As soon as I think to myself ‘I’m so tired,’ then I am instantaneously so much more pumped. The trick is to distract your mind. Think of the sequences in front of you, sing a song to yourself, or keep a mental rhythm going — a lot of methods work but through personal experience, telling yourself that you are pumped or that the next move is hard while you are on the wall definitely does not help. Instead, by focusing your attention on the sole movement in front of you, you can breathe easier, maintain attention, and climb much more precisely.”

— Sasha DiGiulian¹



Mental focus can make or break challenging performances. Clay Cahoon keeps his eye on the prize on *Hellion*, 5.13b, Ten Sleep, WY. © Andrew Burr

FOCUS AND INTENSITY DURING PERFORMANCE

Finally, after countless hours of preparation, the moment of truth has arrived. When it comes time to attempt the redpoint of a limit route, many of the aspects that affect success (physical strength, body weight, technical ability, environmental conditions, etc.) have already been determined. What remains is to summon maximum focus and effort, and then sustain them throughout the duration of the effort.

The first order of business is to ensure that the playing field is properly prepared. This will reduce the chances of some minor detail distracting your attention from the ensuing performance. Are all the quick-draws in place? If not, what gear is required and how should it be racked? Verify that the rock is dry and free of other obstacles. If not, develop a plan for dealing

with the annoyance so it's not a surprise when encountered on route. If stick-clipping, lay out the rope bag first thing, rig the rope through the stick, tag the clip, and then discuss your climbing strategy with your belayer, including:

- Rest locations and durations
- Potential fall locations
- Strenuous clips, runouts, or other locations where the belayer's attention is especially critical
- Locations where extra slack is needed (long dynos or reachy clips)
- Points where verbal encouragement or technical reminders are desired

Once all the trivialities are dealt with, it's time to get "in the zone" for the ensuing performance.

Manifesting intense focus is not trivial, but it does become easier with practice. Fortunately, the weeks of training leading to this point should have provided ample experience with focusing. Each climber will have a slightly different process for getting into the proper *zone* (the mental state of focus and concentration required for optimal performance), and many climbers have different ideas on what that zone should be. For example, some folks prefer complete silence, while others want their mates shouting encouragement. Each individual should determine the best zone for himself, by recalling moments in his past when everything clicked and he climbed confidently. What attitudes and thoughts were present during those moments, and what preparation led to them? Often these factors are difficult to control, but here are some effective methods for entering the zone:

- **Breathing.** Breathing is the key to the mind-body connection, in that it is a subconscious activity that can be controlled consciously.³ Utilize this link to keep the mind calm and attentive, and the body relaxed and supple. Deep breathing will also reduce the likelihood of over-gripping during the climb, since it's difficult to squeeze really hard while breathing from the diaphragm. Start the breathing routine while booting up and tying into the rope. It's much easier to maintain good breathing once it is already started, not to mention that it helps to initiate any strenuous activity with fully oxygenated blood. Rhythmic breathing during final preparation helps calm the nerves, and sends a signal to the belayer that the climber is entering her zone. Breathe in through the nose so that the belly expands completely and causes the chest to rise. Hold the breath for a split-second, and then audibly

exhale slowly and smoothly through the mouth while listening for a consistent rhythm. Effective deep breathing will slow your heart rate and relieve pre-climb jitters.

- **Be Consistent.** The order of final preparation is somewhat arbitrary, but each climber should keep that order consistent, and try to get deeper and deeper into the zone with each step. Simple rituals can be relaxing and put the climber's mind at ease, allowing him to focus completely on the task at hand. They also reduce the chances of forgetting a critical step (like tying in!). Identify a point during the preparation where chit-chat with the belayer should cease so you can get down to business.
- **Use Mental Cues.** Review any last-minute beta cues that are key to the ascent. Go over the general strategy for pacing and note any spots that require extra effort or commitment. Reminders like "shift hips to the right before slapping left hand" can be especially helpful. Finally, perform one last round of visualization, miming the moves while looking at the route. Chalk up one last time, check your knot, harness, and belay, and then go for it.

Once on the route, the climber's inner voice should narrate the moves, making sure to point out key reminders (such as hip shifts, etc.). While primarily focusing directly on the details of the moves in front of him, the climber must also regulate pace, ensure steady breathing, and avoid over-gripping. Fixed mental break points (like good rests or clipping stances) should be used to re-establish a deep, relaxed rhythm. Habitually reset breathing and relaxation each time such points are reached, even during route rehearsal.

Vision can influence thought patterns, so keep your eyes directed on objects that will reinforce mental focus. Depth perception should be limited to seeing no more than five to ten feet ahead, unless you're at a rest. Even at good rests, keep your eyes on the route, and only on the route. Visualize an upcoming crux or recall the locations of important holds. Often during a redpoint or onsight ascent, a looming roof or other distinct crux can be distracting. Obsessing over that particular section won't help you fire the slab 30 feet below it. Keep your eyes focused on the climbing immediately in front of you. During an onsight it is necessary to plan ahead somewhat, but generally long-range planning should be done from the ground or from a good rest stance.

Looking down much beyond the feet should be avoided. The only thing the climber needs to see when looking down is the last piece of solid pro and an attentive belayer. Everything else is a potential distraction. Granted, it may be prudent to downclimb during a challenging onsight, but at that point, the way up is down.

Verbal comments can be extremely helpful or extremely distracting, depending on the climber. These preferences should be discussed with the belayer, and any assembled onlookers, before leaving the ground. The climber shouldn't be afraid to ask for silence if that is his preference (or he can ask his belayer to do so on his behalf once he's left the ground). If verbal feedback is desired, specific cues or reminders should be requested, such as key beta points or reminders to relax and breathe. Ideally these comments should be provided at specific points along the route where the climber is inclined to hold his breath or over-grip. Finally, simple encouragement can be extremely helpful for powerful or committing moves requiring high arousal. Coordinate this encouragement in advance if it is desired.

Remain keenly relaxed throughout the performance. Avoid over-gripping, but be prepared to give maximum effort when the crux is reached. Have confidence that success is possible. If the preparation was thorough, there is a fighting chance. Believe in the beta that was so meticulously developed, believe in the plan, and believe in the training. Maintain an optimal pace, taking care not to rush intricate sequences. Stick to the sequences that were prepared. Rarely do spontaneous sequence changes pan out. There will be ample time to try all sorts of alternate sequences if the planned beta doesn't work out.

Expect the cruxes to feel hard. Don't hesitate unnecessarily. Instead, the climber should move steadily toward the crux and allow his momentum to carry him through. Remain mentally calm even if a desperate effort is required. Once past the crux, attempt to return to a relaxed physical state as quickly as possible by re-enforcing (or reinstating) proper deep-breathing habits and consciously relaxing the grip. Don't act surprised to have made it through the crux. That was the plan all along, so make sure the plan includes confident, cautious optimism once the crux is passed.

Smarter, Not Harder

One of my favorite expressions is "smarter, not harder." Advice to the effect of *try hard* is often misinterpreted to mean *squeeze as hard as possible as you scream like Sharma WHILE YOU ARE CLIMBING!!!* However, most of the time, it turns out that squeezing harder is not the answer. Discovering a new body position, distributing weight perfectly between fingers and toes, rehearsing the accuracy and timing of dynamic movements, pacing the effort correctly — these are often the keys to overcoming a challenging route. Rather than trying hard to execute a move by applying 100 percent of your crimping power, try hard to find a body position that allows you to do the same move with 80 percent exertion!

Furthermore, those that try hard only when they're literally on the rock — climbing — are missing out on countless opportunities to improve their climbing while on terra firma. Try hard to follow your training plan, schedule adequate rest, stick to a healthy diet, and get enough quality sleep. Try hard to wake up on time so that you can get to your project when the conditions are ideal.

At times, *physically* trying hard can be counterproductive, increasing the risk of injury or wearing down skin that could be saved for the next climbing day. Here are some guidelines for working smarter, rather than trying harder:

- Dose your efforts with a preference for quality over quantity
- Listen to your body; if you're too tired to give a quality effort, save it for tomorrow
- Don't climb frustrated, which is a sure path to regret
- Say "Take!" during a too-hard warm-up to avoid a flash pump
- While hangdogging, take a break to tape up before hucking for that sharp, gnarly jug yet again
- Consider new beta before doggedly trying the same move the same way for the tenth time
- Don't feel compelled to perform for the gallery.

When you are on redpoint (or onsight), on a meaningful climb, that is the time to go for it. If you're already in dogging mode, don't be a hero. Use that seemingly useless blob of ballast above your shoulders to do what makes sense in the long run.

— Mark Anderson

It's doubtful everything will unfold flawlessly, so be prepared to overcome some adversity during the ascent. Do not say "Take!" at the first sign of trouble. Keep fighting to the next clipping stance or jug and attempt to regroup by returning to deep, steady breathing and a relaxed mental and physical state. Remain focused on the moves immediately at hand, rather than on anxiety regarding success or failure. If the climber follows through, stays focused, and gives each attempt an honest effort, chances are he will come out on top eventually.

PERFORMANCE REVIEW

After any difficult performance, whether it results in a send or not, the climber should spend a few minutes debriefing the effort (preferably with the belayer or a coach). This is an opportunity to learn and increase the odds of success in the future. Discuss the aspects of the ascent that went right or wrong. The "Performance Review" sidebar recommends a few questions worth considering after any performance event. One purpose of this review is to identify possible triggers that brought about a lack of confidence, lack of resolve, or otherwise prevented you from focusing on the important tasks of breathing properly, climbing efficiently, and executing the plan. Once these triggers are identified, active measures can be used to overcome their influence. A great example of a trigger is dynamic movement, which causes many climbers to hold their breath. With practice and conscious effort, climbers can learn to restart breathing at the conclusion of such moves. Each climber must seek out these triggers in his or her own climbing, and develop a plan to overcome them.

COOLDOWN

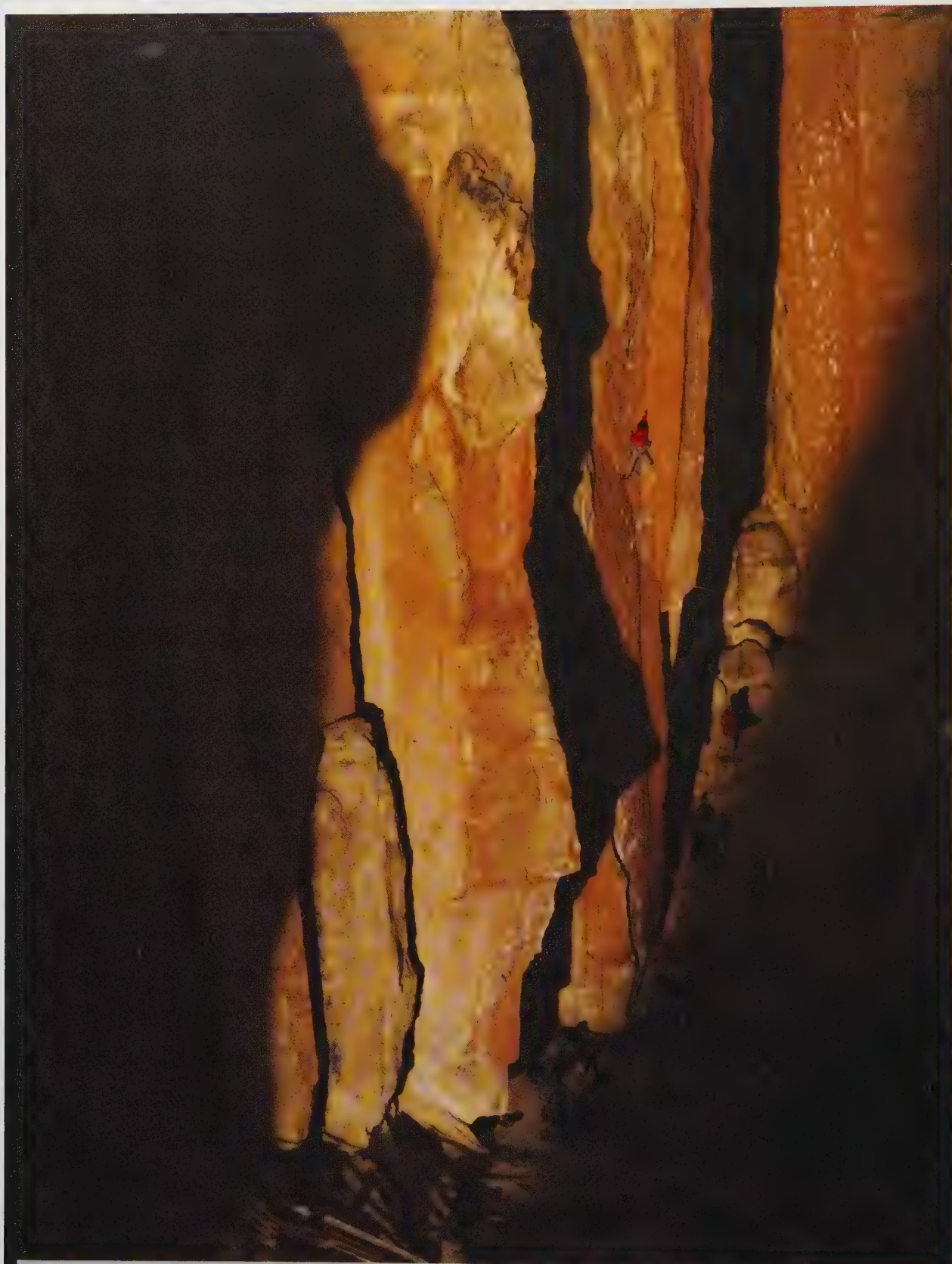
Throughout the climbing day, encourage quick and thorough recovery by refueling and hydrating regularly. Food should be consumed in small, regular doses rather than singular heaping meals, so nibble on energy and protein bars every hour and save the sub sandwich for dinner. Pace climbing effort over the course of the day, limiting the length of working burns to avoid excessive fatigue. While resting between burns, stay out of the sun. Once the hard climbing is done for the day, and the performances have been analyzed for lessons learned, perform a cooldown to hasten the recovery process for the next climbing day. Climb one or two easy yet slightly pumpy routes, which will force blood through the forearms, flushing waste products and delivering blood glucose to the muscles.

Performance Review

Here are some key questions to consider when conducting a post-performance review:

- What went right or wrong?
- Was the warm-up adequate?
- Was there a point at which you started holding your breath or over-gripping?
- Did you maintain a positive attitude?
 - If not, at what point did it break down, and why?
 - If the breakdown was avoided, how did you do it? (Perhaps it was a shout from the belayer to "breathe deep," or a drill used in training that reminded you to relax your grip at the pivotal moment.)
 - Can something be learned from the tenor of your "Inner Dialogue"? Did it start out positive and devolve at a certain point?
- Did you stick to the plan? If not, why not?
- Did you attempt crux moves differently from how you rehearsed them, or did rehearsed sequences feel untenable when on red-point?
- How did cumulative fatigue affect the performance?
- How did you feel physically (lethargic or energetic, strong or weak)?
- How did the committing (dynamic or scary) moves go?

The mind needs a cooldown as well. Most red-point burns end in defeat, so finishing off with a couple laps on a favorite moderate or an easy onsight ends the day on a high note and instills positive memories of the crag and the day. After climbing, thoroughly stretch the forearm flexors and extensors as described in **Chapter 9: Rest, Injury Prevention, and Rehabilitation**. If practical, ice the forearms after any climbing session that causes a deep pump, in order to continue the muscle-flushing process throughout the rest period (a cool stream can be used when camping). See **Chapter 9** for details on icing.



End each crag day with one or more moderate cooldown pitches, like *Moondance*, 5.11b, at Smith Rock, OR.  John Borland www.morffed.com

SUMMARY

- External factors can make or break climbing performances.
- Be cognizant of environmental conditions affecting your goal routes, and plan your visits to coincide with periods of stable, predictable weather.
- Climb in the shade whenever possible.
- Ensure that you're familiar with your climbing equipment, that it inspires confidence, and that it is in good working order.
- Be creative and exhaustive when discovering beta. Consider every possible option, and ask your partner to participate in this process. Consider clip and rest stances as well.
- Document your chosen sequence on video, paper, or both.
- Visualization is a powerful tool for improving performance. Visualize your sequence several times per day when working a difficult project.
- A thorough warm-up is essential for optimal performance. Warm-up the body *and* the mind.
- Stay focused during the performance. Practice getting in "the zone," use good breathing habits, and keep your attention on the task at hand.
- A broad and thorough route pyramid will provide the needed confidence to believe you will succeed on your goal route.
- Review each performance with your partner to identify any opportunities for improvement.
- End each climbing day with a cooldown. A cooldown will speed recovery between climbing days and ensure that each day ends on a high note.

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